

CMR INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
PROJECT PHASE 1 (BAD685) – SYNOPSIS

1. Title of the Project

CampusQuant AI: Data-Driven Talent Forecasting and Placement Intelligence System

2. Introduction / Background

Campus placements are one of the most important aspects of a student's academic journey, as they directly impact career opportunities after graduation. In many colleges, the placement process is still managed using basic portals or manual methods, which mainly focus on student registration and company job postings. These systems do not provide intelligent insights into student readiness, placement trends, or skill gaps.

With the increasing demand for data-driven recruitment and personalized student preparation, there is a need for a smarter placement support system. The proposed project, CampusQuant AI: Data-Driven Talent Forecasting and Placement Intelligence System, aims to address this need by integrating student data, company recruitment data, and AI-based analytics.

The system will help students assess their placement readiness, identify skill gaps, and receive role-based preparation support. It will also assist placement officers in analyzing placement trends, skill demand, and batch-level readiness, while allowing recruiters to manage job postings and candidate shortlisting efficiently. Thus, the project provides a more intelligent and practical solution compared to traditional placement management systems.

3. Problem Statement

In many colleges, the placement process is handled through basic management systems or manual records that only store student and company information. These systems lack intelligent analysis and decision support for both students and placement officers.

The limitations of existing systems include:

- No proper analysis of student placement readiness
- No automated eligibility filtering based on company requirements
- No skill gap detection for students
- Lack of data-driven placement forecasting
- Limited interaction between placement officers and recruiters
- No intelligent recommendation of job roles or preparation roadmap

As a result, students may not be fully prepared for placement drives, placement officers may find it difficult to plan training activities, and recruiters may not efficiently identify suitable candidates.

Therefore, there is a need for an AI-based placement intelligence system that can improve placement preparation, automate recruitment workflows, and provide data-driven insights for better placement outcomes.

4. Objectives of the Project

- To design and develop a web-based placement intelligence platform for students, placement officers, and recruiters
- To create a student profile management system using resume upload and manual data entry
- To analyze student resumes using AI/NLP-based resume analysis
- To evaluate student performance using aptitude, coding, and soft skills levels
- To predict student placement readiness using machine learning models
- To identify student skill gaps and provide training recommendations

- To recommend suitable job roles and companies based on student profile data
- To automate eligibility filtering and candidate shortlisting using company recruitment criteria
- To provide interactive placement analytics dashboards for placement officers
- To forecast batch-level placement outcomes using historical and current student data
- To support continuous updates by placement officers and recruiters for dynamic placement management

5. Expected Outcome / Results

- A dynamic web application for campus placement management
- A student portal that supports resume analysis, placement readiness evaluation, skill gap analysis, and role recommendations
- An admin / placement officer dashboard for student monitoring, company management, placement statistics, and placement forecasting
- A recruiter portal for job posting, candidate filtering, shortlisting, interview evaluation, and offer letter management
- A machine learning model that predicts student placement probability using academic and skill-based inputs
- Interactive dashboards for placement trends, skill demand analysis, and recruitment analytics
- A system that improves the efficiency of the placement process and supports data-driven decision making

6. Applications

- College placement cells for managing campus recruitment activities
- Engineering and degree colleges for student placement readiness tracking
- Training and placement departments for identifying skill gaps and planning training programs
- Campus recruitment drives for company-wise eligibility filtering and candidate shortlisting
- Academic institutions for placement analytics and reporting
- Future extension into internship recommendation and career guidance systems